Ex no:

BOOK BANK SYSTEM

Date:

AIM:

To create a system to perform book bank operation

(I) PROBLEM STATEMENT:

A Book Bank lends books and magazines to member, who is registered in the system. Also it handles the purchase of new titles for the Book Bank. Popular titles are brought into multiple copies. Old books and magazines are removed when they are out or date or poor in condition. A member can reserve a book or magazine that is not currently available in the book bank, so that when it is returned or purchased by the book bank, that person is notified. The book bank can easily create, replace and delete information about the tiles, members, loans and reservations from the system.

(II) SOFTWARE REQUIREMENTS SPECIFICATION:

1.0 INTRODUCTION

Book Bank is the interface between the students and Librarian. It aims at improving the efficiency in the Issue of books or magazines and reduce the complexities involved in it to the maximum possible extent.

1.1 PURPOSE

If the entire process of 'Issue of Books or Magazines' is done in a manual manner then it would take several months for the books or magazines to reach the applicant. Considering the fact that the number of students for Book Bank is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process. The system has been carefully verified and validated in order to satisfy it.

1**.2 SCOPE**

The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning). The authority concerned with the issue of books can use this system to reduce his workload and process the application in a speedy manner.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

- **Librarian** - Refers to the super user who is the Central Authority who has been vested with the privilege to manage the entire system.
- **Student** - One who wishes to obtain the Books or Magazines.
- **HTML** - Markup Language used for creating web pages.

- **J2EE** - Java 2 Enterprise Edition is a programming platform and it is the part of the java platform for developing and running distributed java applications.
- **HTTP** -Hyper Text Transfer Protocol
- **TCP/IP** - Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format

1.5 TECHNOLOGIES TO BE USED

- Visual Basic
- Oracle 11g

1.6 TOOLS TO BE USED

- Visual Basic Tools
- Rational Rose tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements.

Overall description will describe major role of the system components and inter-connections.

Specific requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION:

It will describe major role of the system components and inter-connections.

2.1 PRODUCT PERSPECTIVE

The SRS acts as an interface between the 'Students' and the 'Librarian'. This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration in which the user receives the books or magazines.

2.2 SOFTWARE INTERFACE

- **Front End Client** - The Student and Librarian online interface is built using Visual studio.
- **Back End** - Oracle 11 g database

2.3 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

- Secure Registration of information by the Students.

- Librarian can generate reports from the information and is the only authorized personnel to add the eligible application information to the database.

2.5 USER CHARACTERISTICS

- **Student** - They are the people who desire to obtain the books and submit the information to the database.
- **Librarian** - He has the certain privileges to add the books and to approval of the reservation of books.

2.6 CONSTRAINTS

- The Students require a computer to submit their information.
- Although the security is given high importance, there is always a chance of intrusion in the web world which requires constant monitoring.
- The Students has to be careful while submitting the information. Much care is required.

2.7 ASSUMPTIONS AND DEPENDENCIES

- The Student and Librarian must have basic knowledge of computers and English Language.
- The Students may be required to scan the documents and send.

(III)USE-CASE DIAGRAM:

The book bank use cases are:

1. book_issue
2. book_return
3. book_order
4. book_entry
5. search book_details

ACTORS INVOLVED:

1. Student
2. Librarian
3. Vendor

USECASE NAME : SEARCH BOOK_DETAILS

The librarian initiates this use case when any member returns or request the book and checking if the book is available.

Precondition: The librarian should enter all Book details.

Normal Flow: Build message for librarian who search the book.

Post Condition: Send message to respective member who reserved the book.

USECASE NAME : BOOK_ISSUE

Initiated by librarian when any member wants to borrow the desired book. If the book is available, the book is issued.

Precondition: Member should be valid member of library.

Normal Flow: Selected book will be issued to the member.

Alternative Flow: If book is not available then reserved book use case should be initiate.

Post Condition: Update the catalogue.

USECASE NAME : BOOK_ORDER

Initiated by librarian when the requested book is not available in the library at that moment. The book is reserved for the future and issued to the person when it is available.

Precondition: Initiated only when book is not available.

Normal Flow: It reserved the book if requested.

Post Condition : Mention the entry in catalogue for reservation.

USECASE NAME : BOOK_RETURN

Invoked by the librarian when a member returns the book.

Precondition: Member should be valid member of library.

Normal Flow: Librarian enters bookid and system checks for return date of the book.

Alternative Flow: System checks for return date and if it returned late fine message will be displayed.

Post Condition: Check the status of reservation.

USECASE NAME : BOOK_ENTRY

The purchase book use-case when new books invoke it or magazines are added to the library.

Precondition: Not available or more copies are required.

Normal Flow: Enter bookid,author information, publication information, purchased date, prize and number of copies.

Post Condition: Update the information in catalogue.

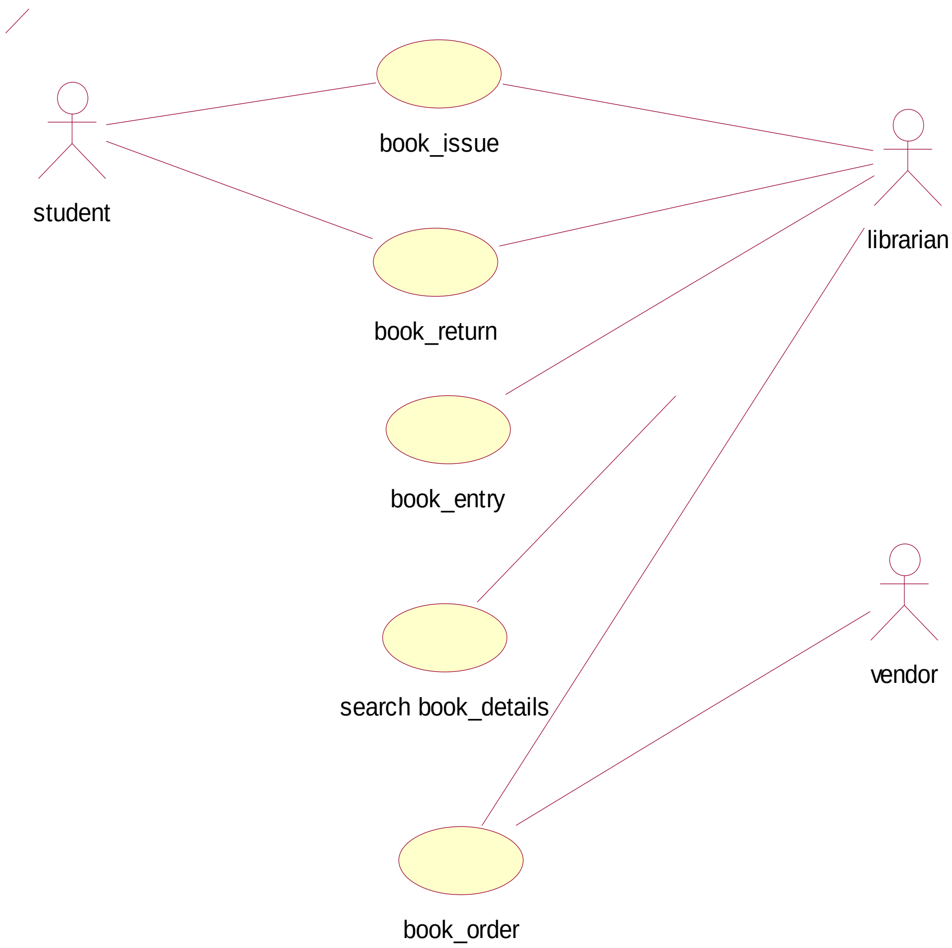


Fig 3.1 USE-CASE DIAGRAM FOR BOOK BANK SYSTEM

(IV) ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control. An activity is shown as an rounded box containing the name of the operation.

This activity diagram describes the behaviour of the system.

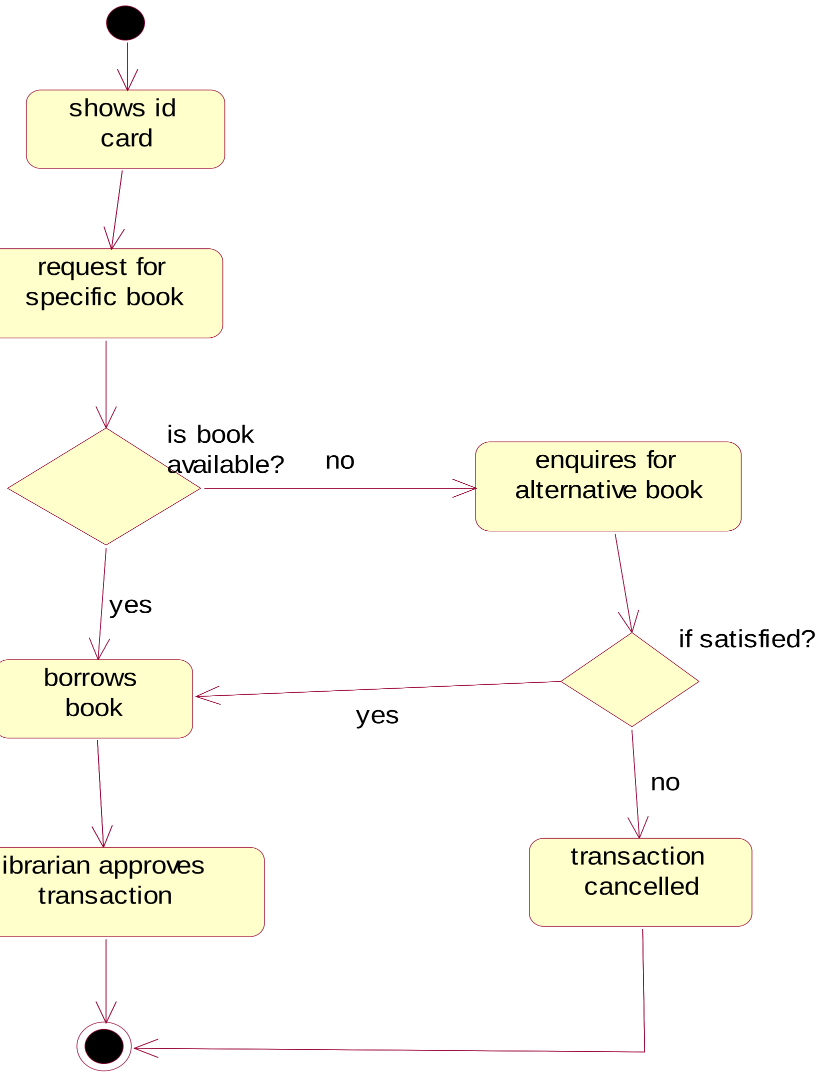


Fig4.1 ACTIVITY DIAGRAM [BORROW BOOK]

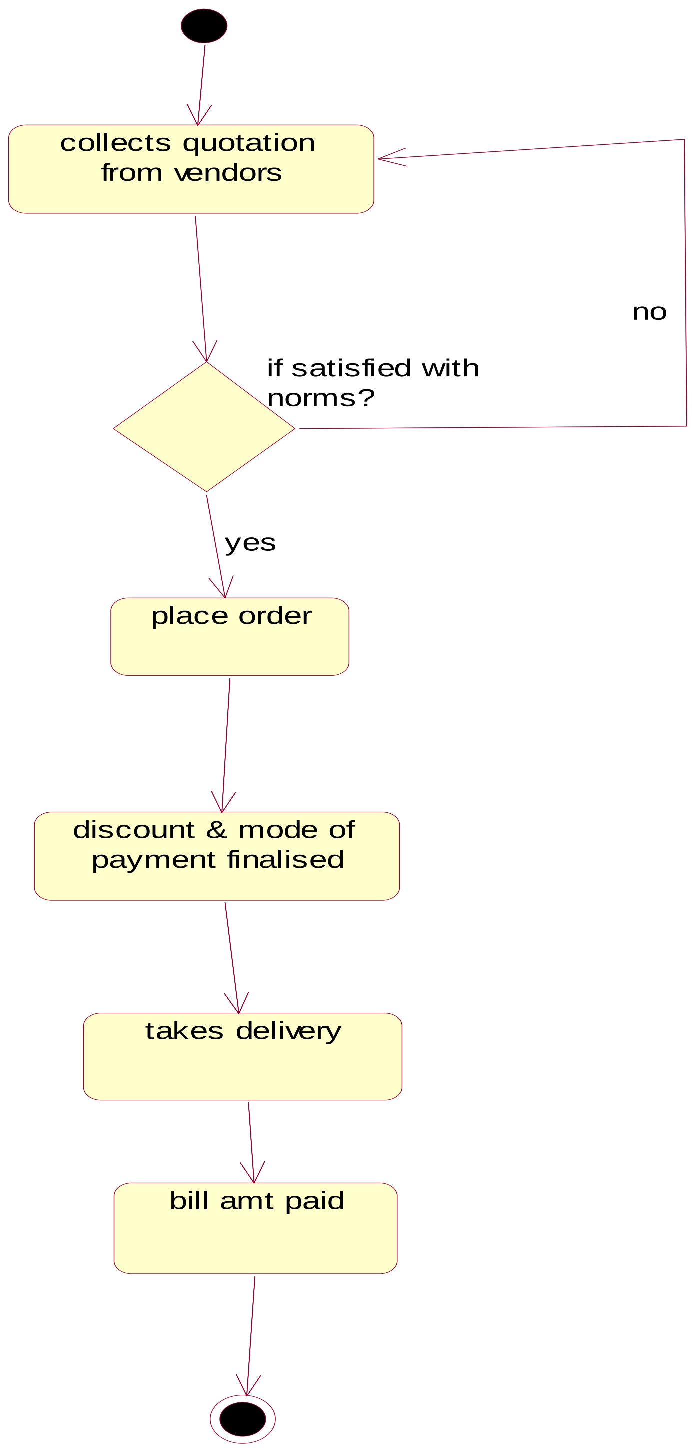
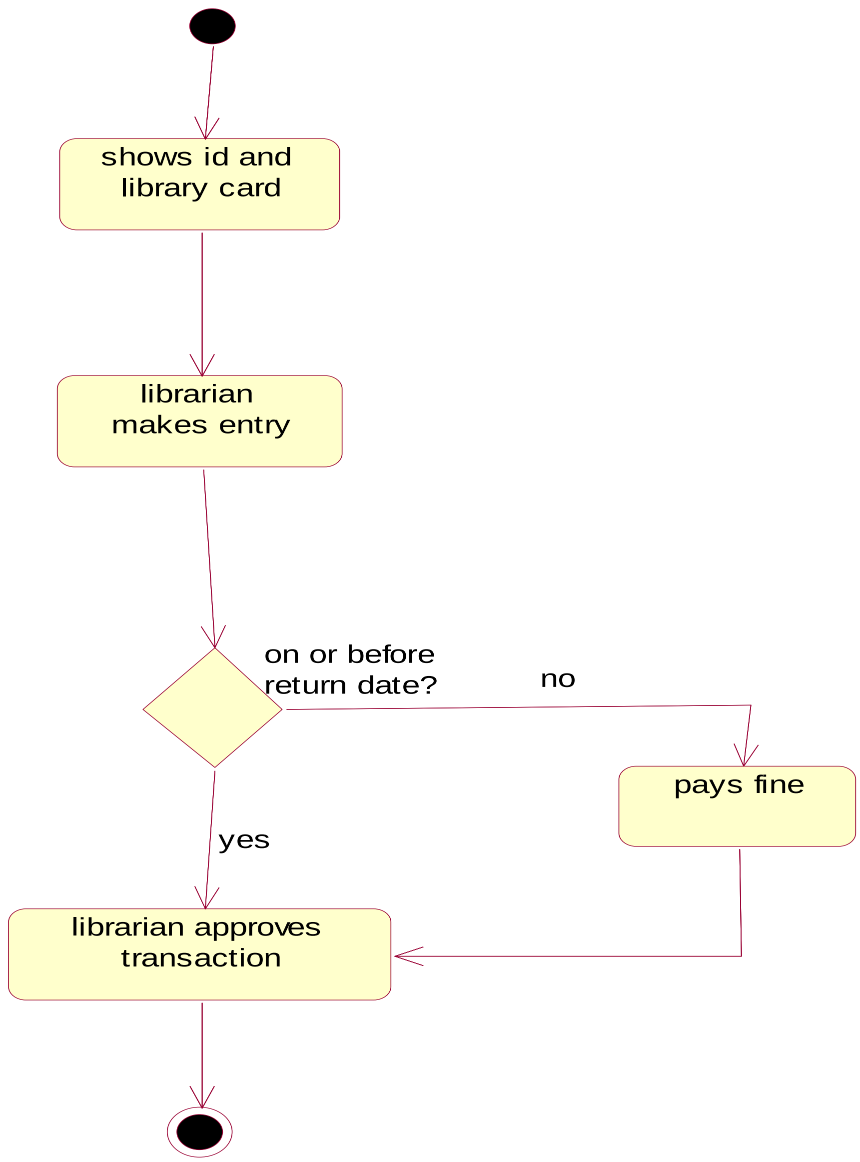


Fig.4.2 ACTIVITY DIAGRAM [ORDER BOOK]

Fig.4. 3ACTIVITY DIAGRAM [RETURN BOOK]



(V) CLASS DIAGRAM:

The class diagram, also referred to as object modeling is the main static analysis diagram. The main task of object modeling is to graphically show what each object will do in the problem domain. The problem domain describes the structure and the relationships among objects.

The ATM system class diagram consists of four classes:

1. Student
2. Book
3. Issue
4. Return
5. Vendor
6. Details

1) STUDENT:

It consists of twelve attributes and three operations. The attributes are enrollno, name, DOB, fathername, address, dept name, batch and book limits. The operations of this class are addStInfo(), deleteStInfo(), modifyStInfo().

2) BOOK:

It consists of ten attributes and four operations. This class is used to keep book information such as author, title, vendor, price, etc

3) ISSUE:

It consists of eight attributes and two operations to maintain issue details such as, issue date, accno of issued book, name of the student who borrowed book.

4) RETURN:

It consists of eight attributes and two operations to maintain issue details such as, issue date, accno of issued book, name of the student who borrowed book.

5) STUDENTS:

The attributes of this class are name, dept ,year ,bcode no The operation is display students().

6) DETAIL:

The attributes of this class are book name, author, bcode no The operations are delete details().

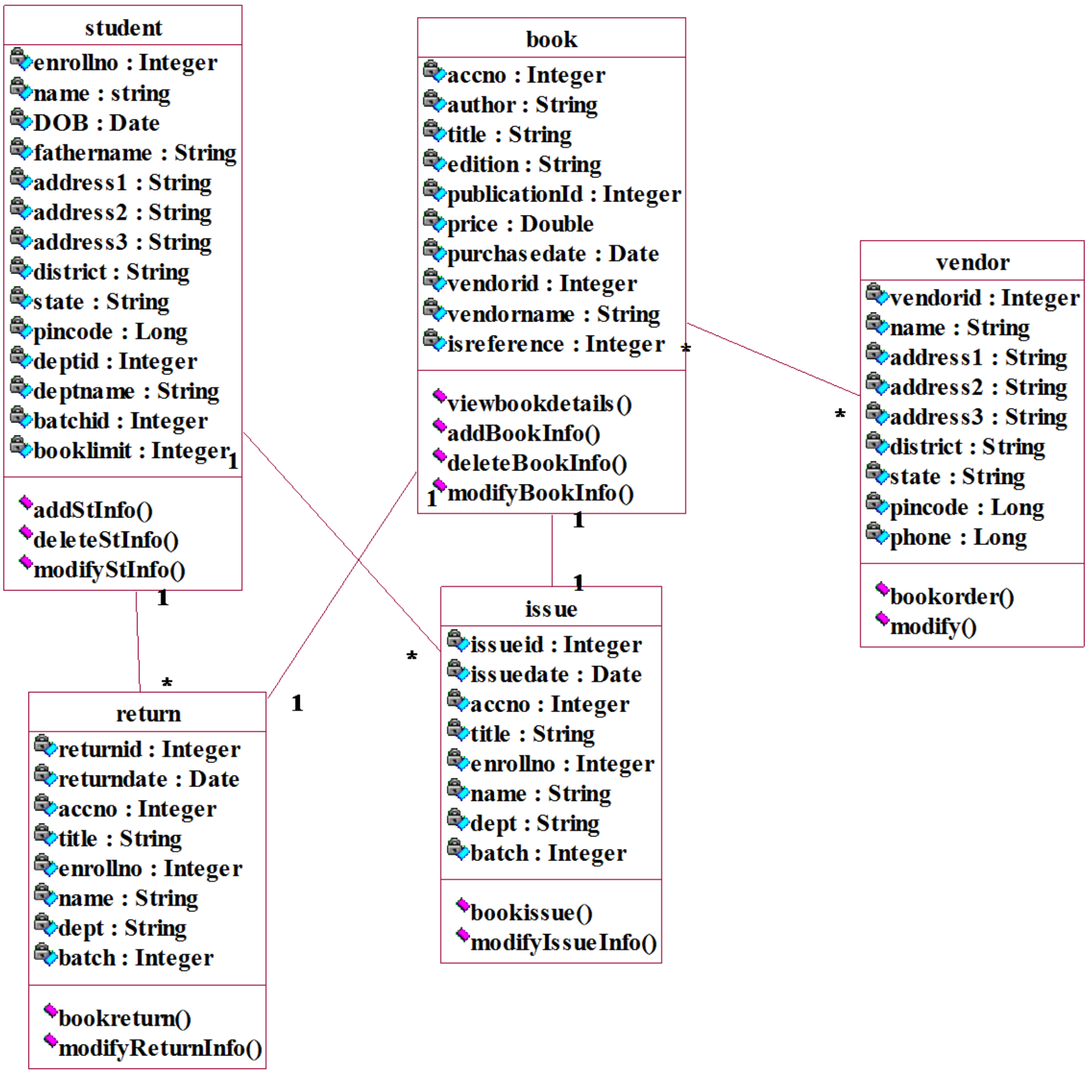


Fig.5. CLASS DIAGRAM FOR BOOK BANK SYSTEM

(VI) SEQUENCE DIAGRAM:

A sequence diagram represents the sequence and interactions of a given USE-CASE or scenario. Sequence diagrams can capture most of the information about the system. Most object to object interactions and operations are considered events and events include signals, inputs, decisions, interrupts, transitions and actions to or from users or external devices.

An event also is considered to be any action by an object that sends information. The event line represents a message sent from one object to another, in which the “form” object is requesting an operation be performed by the “to” object. The “to” object performs the operation using a method that the class contains.

It is also represented by the order in which things occur and how the objects in the system send message to one another.

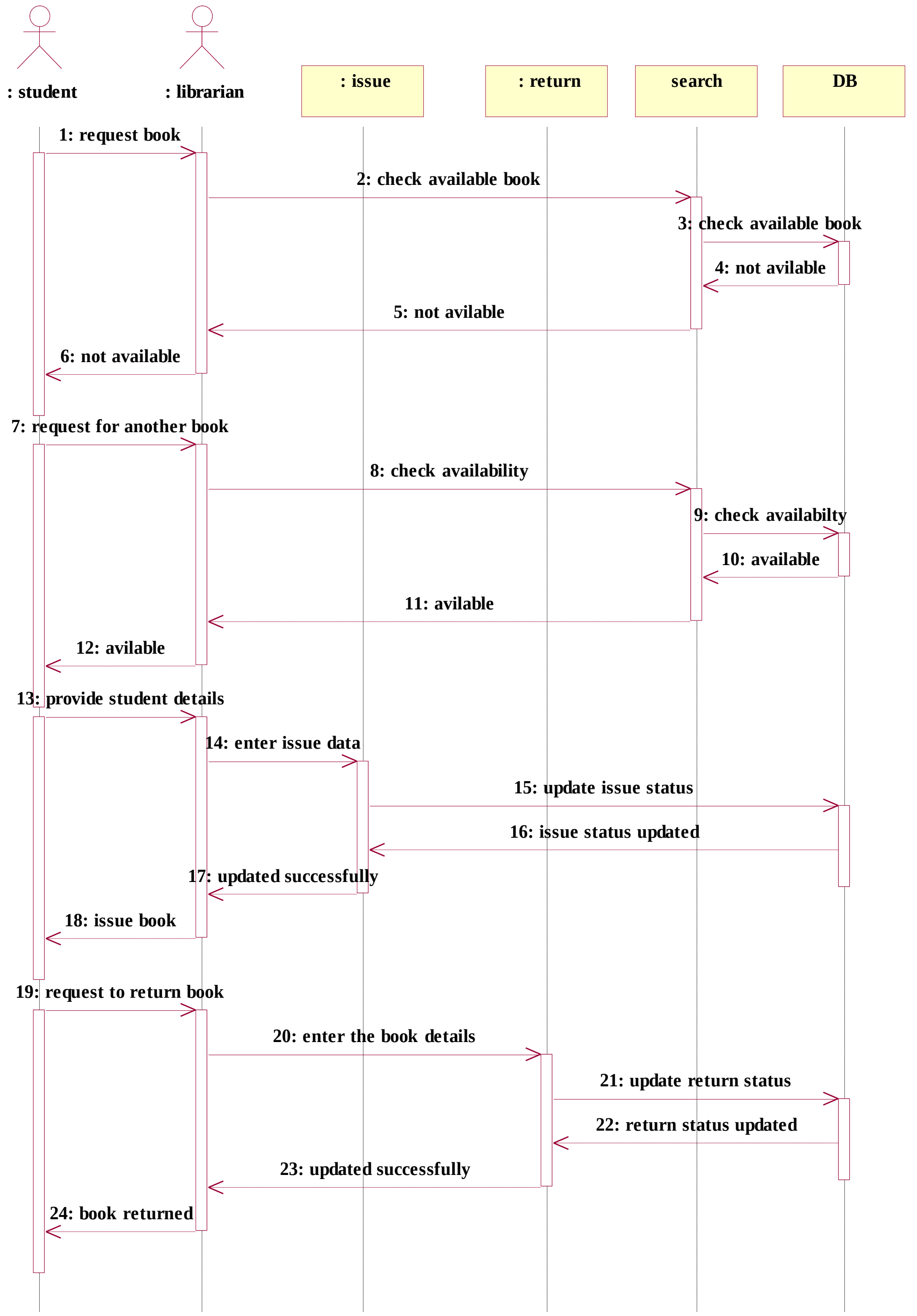


Fig. 6.1. SEQUENCE DIAGRAM FOR BOOK ISSUE & RETURN

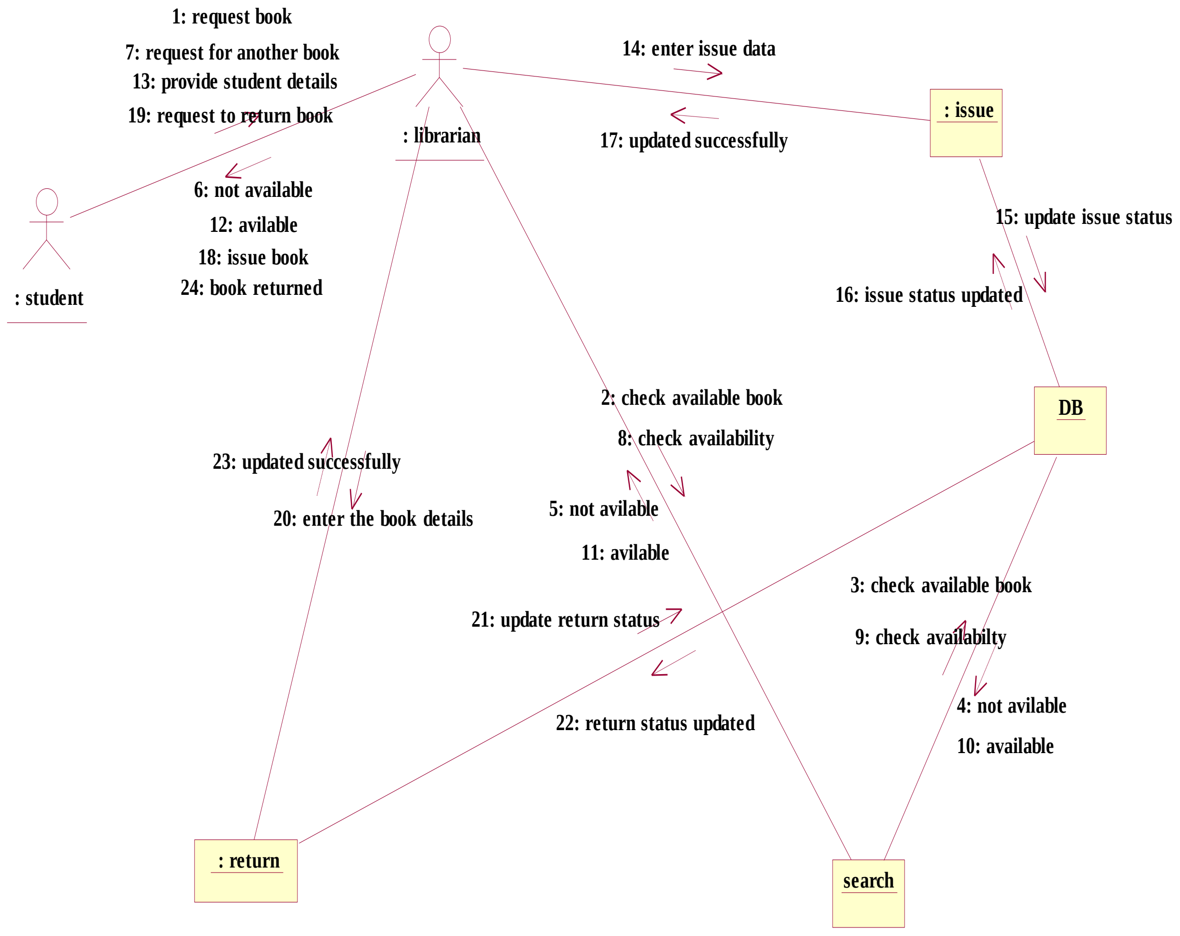


Fig. 6.2. COLLABORATION DIAGRAM FOR BOOK ISSUE & RETURN

(VII) STATE CHART DIAGRAM

It consists of state, events and activities. State diagrams are a familiar technique to describe the behavior of a system. They describe all of the possible states that a particular object can get into and how the object's state changes as a result of events that reach the object

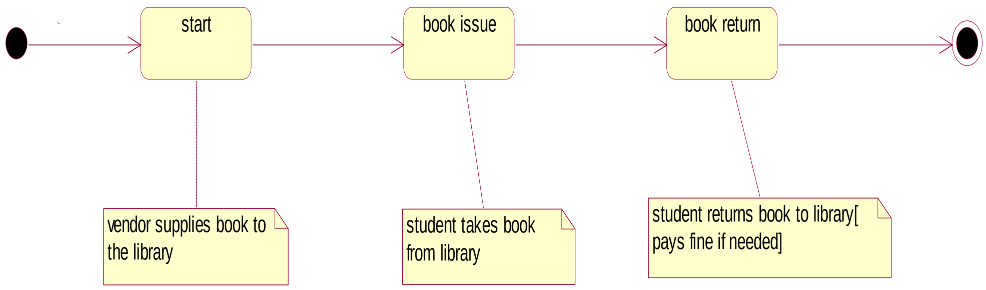


Fig. 7.1 STATE CHART DIAGRAM

(VIII) DEPLOYMENT DIAGRAM AND COMPONENT DIAGRAM

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.

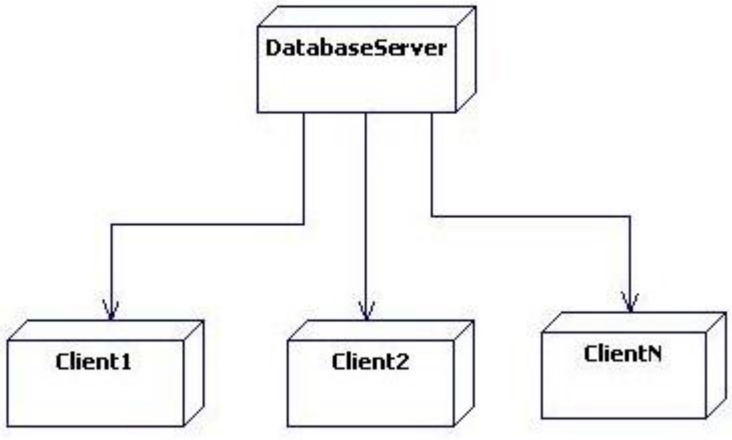


Fig.8.1.DEPLOYMENT DIAGRAM

(IX) IMPLEMENTATION OF DOMAIN OBJECTS LAYER AND TECHNICAL SERVICE LAYER

```
//Source file: E:\\sruthi\\book.java

public class book1
{
   private integer accno;
   private string author;
   private string title;
   private string edition;
   private integer publicationid;
   private double price;
   private date purchaseDate;
   private integer vendorId;
   private string vendorName;
   private integer isReference;
   public return1 theReturn1;
   public vendor1 theVendor1;
```

```
  /**
  @roseuid 51187CA503B2
   */
  public book()
  {

  }

  /**
  @roseuid 510F44A7007E
   */
  public void viewBookDetails()
  {

  }

  /**
  @roseuid 510F44B50217
   */
  public void addbookinfo()
  {

  }

  /**
  @roseuid 510F44BD0063
   */
  public void deletebookinfo()
  {

  }

  /**
  @roseuid 510F44C60045
   */
  public void modifybookinfo()
  {

  }
}
/**

void book1.newbookdetails(){

  }

void book1.modify(){

  }
void book1.add(){
```

```
   }
void book1.viewBookDetails(){

   }
void book1.bookEntry(){

   }
void book1.delete(){

   }

 */
```

//Source file: D:\\vaish\\issue.java

```
public class issue
{
   private date iisDate;
   private integer issId;
   private integer acno;
   private string title;
   private integer enNo;
   private string name;
   private integer deptId;
   private integer batId;
   public books theBooks;

   /**
   @roseuid 5100DE750281
    */
   public issue()
   {

   }
}
//void issue.modify(){
//
//   }
//void issue.bookissue(){
//
//   }
```

//Source file: F:\\VAISH\\return1.java

```
public class return1
{
   public integer returnid;
   public date returndate;
```

```
  public integer accno;
  public string titlle;
  public integer enrollno;
  public string name;
  public string dept;
  public integer batch;
  public book theBook;

  /**
  @roseuid 50F92E4D0177
   */
  public return1()
  {

  }

  /**
  @roseuid 50F91C69005D
   */
  public void bookreturn()
  {

  }

  /**
  @roseuid 50F91C6C0000
   */
  public void modify()
  {

  }
}
```

//Source file: F:\\VAISH\\student.java

```
public class student
{
  public integer enrollno;
  public string name;
  public date DOB;
  public string fathername;
  public string address;
  public string address2;
  public string address3;
  public string district;
  public string state;
  public long pincode;
  public integer deptid;
  public string depname;
```

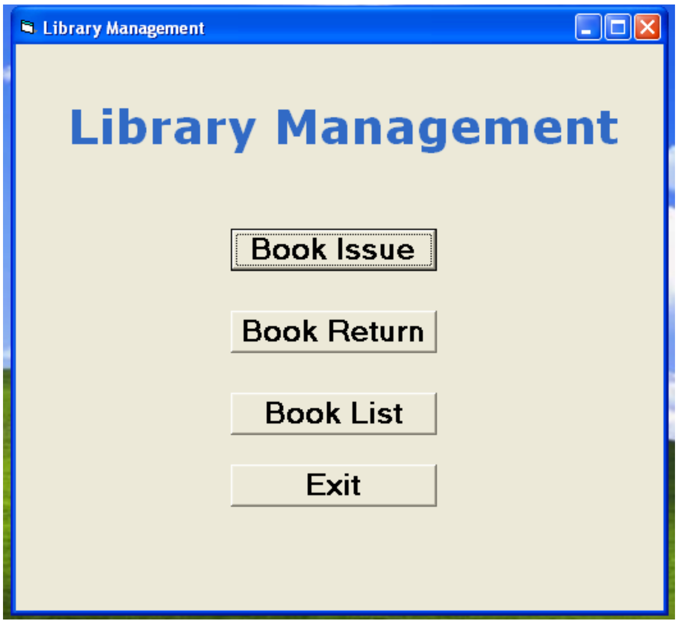
```
   public integer batchid;
   public integer booklimit;
   public issue theIssue;
   public return1 theReturn1;

   /**
   @roseuid 50F92E4D00FA
    */
   public student()
   {

   }

   /**
   @roseuid 50F918E6000F
    */
   public void add()
   {

   }

   /**
   @roseuid 50F918EB03A9
    */
   public void delete()
   {

   }

   /**
   @roseuid 50F918EE029F
    */
   public void modify()
   {

   }
}
```

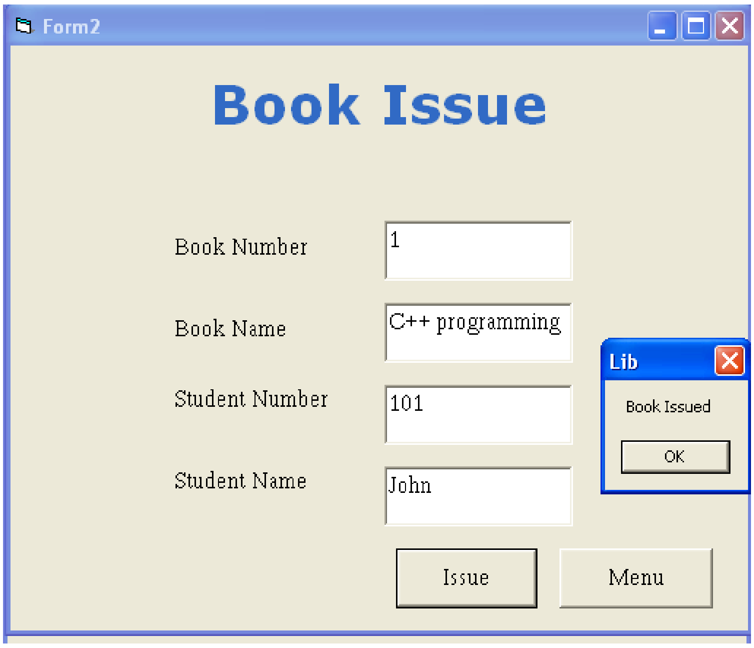
//Source file: F:\\VAISH\\vendor1.java

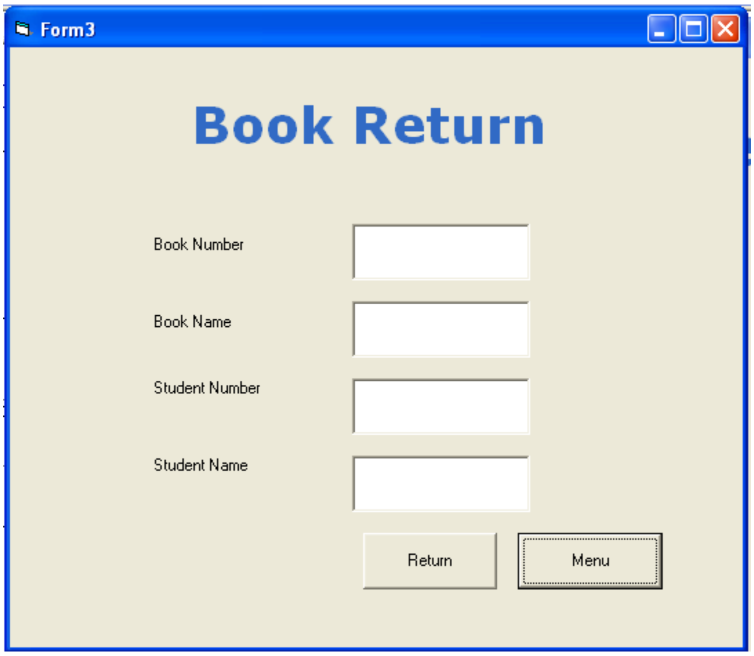
```
public class vendor1
{
   private integer vendorid;
   private string name;
   private string address1;
   private string address2;
   private string address3;

   /**
```

```
  @roseuid 5118C814009C
   */
  public vendor1()
  {

  }

  /**
  @roseuid 5118B75100AB
   */
  public void bookOrder()
  {

  }

  /**
  @roseuid 5118B75702BF
   */
  public void modify()
  {

  }
}
//void vendor1.bookorder(){
//
//   }
```

(X) IMPLEMENTATION OF USER INTERFACE LAYER







RESULT:

Thus the mini project for Book Bank System has been successfully executed and codes are generated.